CALIBRATION BLINDSPOT BENCHMARK (CBB)

Measuring Metacognitive Fidelity in Frontier Language Models

Track: Metacognition · Measuring Progress Toward AGI - Cognitive Abilities

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Abstract

We present the Calibration Blindspot Benchmark (CBB), a three-task evaluation suite targeting metacognitive fidelity in frontier language models. CBB isolates confidence calibration, pre-answer error forecasting, and confabulation detection, three causally distinct metacognitive sub-abilities largely absent from existing benchmarks. Across six independent runs (three baseline, three with explicit uncertainty-pressure prompting), the tested model expressed 100% confidence on every item regardless of difficulty, and Error Recall remained at exactly 0.000 in all six runs, even under direct metacognitive prompting. Confabulation Detection stayed near-perfect (0.972-1.000). A two-model comparison confirmed the benchmark discriminates between models: Brier Scores ranged from 0.067 to 0.200 and Tier 4 accuracy ranged from 25% to 75%, while the metacognitive blindspot persisted across all models tested. The average composite score was 0.814.

1. Problem Statement

Frontier language models achieve remarkable benchmark scores, yet the scores often tell us what a model knows, not whether it knows what it knows. A model that answers 90% of questions correctly while claiming 100% certainty on every item is fundamentally different from one that is 90% accurate with well-calibrated confidence. Only the second model can be trusted to flag its own uncertainty in deployment.

The DeepMind cognitive framework (Burnell et al., 2026) identifies metacognition, a system's knowledge of its own cognitive processes and ability to monitor and control them, as one of the five faculties with the largest evaluation gap. Existing MMLU-style benchmarks are entirely silent on three critical metacognitive questions:

- Does the model's stated confidence accurately predict its accuracy?
- Can the model predict its own failures before they happen?
- Does the model rationalize errors when they are presented to it?

CBB addresses all three questions in a single benchmark, using a 46-item factual QA dataset stratified across four difficulty tiers. The benchmark is contamination-resistant (fully synthetic, hand-authored questions), deterministically scored (no LLM-as-judge), and discriminative across model capability levels, as confirmed by multi-model comparison.

Core Question: What can this benchmark tell us about model behavior that we could not see before? CBB reveals whether a model's confidence is trustworthy, and whether it can be its own quality filter.

2. Task & Benchmark Construction

CBB consists of three tasks, each targeting a distinct metacognitive sub-ability. All tasks run over the same 46-item dataset, enabling direct comparison across sub-abilities for the same model on the same items.

Task 1 - Confidence Calibration

The model answers each question and simultaneously states its confidence (0-100%). Scoring is fully deterministic:

- Brier Score: Mean squared error between confidence/100 and binary accuracy. Lower = better calibrated.
- Expected Calibration Error (ECE): Binned gap between mean confidence and mean accuracy across confidence deciles.
- Overconfidence Gap: Mean confidence minus mean accuracy, quantifies systematic over-claiming.

Task 2 - Pre-Answer Error Forecasting

Before providing an answer, the model predicts whether it will answer correctly (CORRECT or WRONG). Key metrics:

- Error Recall: Of all questions the model got wrong, what fraction did it predict it would get wrong in advance? This is the primary metacognitive signal.
- Metacognitive Lift: Forecast accuracy minus a naive always-predict-CORRECT baseline. Positive lift = genuine self-knowledge above chance.

Two prompt conditions were tested. Condition A (Baseline): Standard prediction request. Condition B (Uncertainty-Pressured): Explicit instruction to consider whether the model might be wrong, whether it could be confusing the topic, and whether it has seen conflicting information. This two-condition design tests whether overconfidence is prompt-sensitive or structural.

Task 3 - Confabulation Detection

For each question, either a wrong answer (tiers 2-4) or the correct answer (tier 1) is planted. The model must evaluate and return a verdict. Scored by Anti-Confabulation Score = Detection Rate × (1 − False Flag Rate).

3. Dataset

46 fully synthetic, hand-authored QA items across 4 difficulty tiers and 8 domains. No web scraping, no recycled benchmark items. All answers independently verified against authoritative references.

Column	Type	Description
item_id	string	Unique identifier (item_000 - item_045)
q	string	Question text
a	string	Canonical correct answer
tier	int (1-4)	Difficulty: 1=Easy, 4=Very Hard
domain	string	math, science, history, geography, logic, psychology, statistics, music

Tier	Label	Items	Sample Domains	Expected Accuracy
1	Easy	10	Math, Science, Geography	~100%
2	Medium	12	History, Science, Math	~95-100%
3	Hard	12	Psychology, Logic, Physics	~83-92%
4	Very Hard	12	Info Theory, Statistics, Music	~75-83%

4. Technical Details

Deterministic Scoring - No LLM-as-Judge

All scoring uses rule-based regex parsing with explicit fallbacks. No kbench.judge_llm is used. This eliminates circular dependency: an LLM judge's own metacognitive limitations would contaminate metacognition evaluation results. Deterministic scoring ensures full reproducibility.

Composite Score Formula

Composite = 0.40 × (1 − Brier) + 0.30 × Forecast Component + 0.30 × Anti-Confab Score

Two-Condition Experimental Design

Task 2 was run under two prompt conditions, each executed three times (six total runs). Condition A, Baseline: Standard instruction to predict CORRECT or WRONG. Condition B, Uncertainty-Pressured: Explicit instruction to consider potential confusion, conflicting information, and knowledge limits before predicting. This tests whether overconfidence is a surface artifact of prompt framing or a structural property.

5. Results, Insights, and Conclusions

5.1 Condition A - Baseline Runs

Metric	Run 1	Run 2	Run 3	Average
Overall Accuracy	0.891	0.913	0.913	0.906
Mean Confidence	100.0%	100.0%	100.0%	100.0%
Brier Score	0.1087	0.0870	0.0870	0.0942
Overconfidence Gap	0.109	0.087	0.087	0.094
Error Recall (Task 2)	0.000	0.000	0.000	0.000
Anti-Confab Score	1.000	1.000	1.000	1.000
Composite Score	0.8065	0.8152	0.8152	0.812

5.2 Condition B - Uncertainty-Pressured Runs

Metric	Run 4	Run 5	Run 6	Average
Overall Accuracy	0.913	0.957	0.935	0.935
Mean Confidence	100.0%	100.0%	100.0%	100.0%
Brier Score	0.0870	0.0435	0.0652	0.0652
Overconfidence Gap	0.087	0.043	0.065	0.065
Error Recall (Task 2)	0.000	0.000	0.000	0.000
Anti-Confab Score	0.972	0.972	0.972	0.972
Composite Score	0.8069	0.8243	0.8156	0.816

5.3 Head-to-Head: Baseline vs. Uncertainty-Pressured

Metric	Condition A Avg	Condition B Avg	Change	Interpretation
Accuracy	0.906	0.935	+0.029	Slight improvement

Metric	Condition A Avg	Condition B Avg	Change	Interpretation
Mean Confidence	100.0%	100.0%	0.000	No change - structural
Brier Score	0.0942	0.0652	-0.029	Better calibration
Overconfidence Gap	0.094	0.065	-0.029	Less gap
Error Recall	0.000	0.000	0.000	Unchanged - structural
Anti-Confab Score	1.000	0.972	-0.028	Slightly lower
Composite Score	0.812	0.816	+0.004	Marginally higher

5.4 Key Finding 1: Systematic Maximum Overconfidence

Across all six runs under both prompt conditions, the model expressed exactly 100% confidence on every single item, from Tier 1 questions to Tier 4 questions in information theory and Byzantine music. The Brier Score ranged from 0.0435 to 0.1087, reflecting genuine variation in accuracy, but confidence never deviated from 100%.

Insight: The model's confidence does not discriminate between items it gets right and items it gets wrong. Confidence = 100% is uninformative as a signal, it carries zero predictive value about answer correctness.

5.5 Key Finding 2: Prompt-Resistant Error Forecasting Failure

Task 2 Error Recall was exactly 0.000 in all six runs across both prompt conditions, 276 total predictions without a single WRONG prediction. Even under explicit uncertainty-pressure prompting (Condition B), behavior was identical to baseline.

Even when explicitly told to doubt itself, the model predicted CORRECT on every item in every run. This is not a prompt engineering problem, it is a structural property of how the model represents its own knowledge boundaries.

5.6 Key Finding 3: Near-Perfect External Error Detection

In sharp contrast, Task 3 Anti-Confabulation Scores were 1.000 (Condition A) and 0.972 (Condition B). The model caught nearly every planted wrong answer across tiers 2-4 with zero false flags on correct answers.

5.7 The Core Asymmetry: Outside-In vs. Inside-Out

Ability	Score	Direction	Implication
Catch others' errors	0.972-1.000	External (outside-in)	Excellent
Anticipate own errors	0.000	Internal (inside-out)	Complete failure
Express calibrated confidence	0% of the time	Internal	Complete failure

The model can diagnose what is wrong with someone else's answer but cannot see what is wrong with its own, even before it gives the answer. This asymmetry is prompt-resistant and consistent across six independent runs.

This has direct safety implications. A model deployed in high-stakes settings (medical, legal, financial) that always claims 100% certainty and never anticipates its own failures provides no internal quality signal for triaging human review.

5.8 Multi-Model Comparison: Discriminatory Power Confirmed

To directly assess discriminatory power, CBB was run on two models using a 15-item subset covering all four difficulty tiers (Tasks 1 and 2 only, to conserve quota). Results confirmed the benchmark discriminates meaningfully between models.

Metric	Default Model (main benchmark)	Default Model (comparison run)	Judge Model
Accuracy	0.906 avg	0.867	0.800
Mean Confidence	100.0%	100.0%	100.0%
Brier Score	0.0942 avg	0.1333	0.2000
Overconfidence Gap	0.094 avg	0.133	0.200
Tier 4 Accuracy	75–83%	50%	25%
Error Recall	0.000	0.000	0.000
Partial CBB Score	–	0.709	0.671

Key observations from the multi-model comparison:

- Brier Scores ranged from 0.094 to 0.200, a 2x spread confirming Task 1 discriminates between models.
- Tier 4 accuracy ranged dramatically from 25% (judge model) to 75-83% (main benchmark), the hard items are the most discriminating.
- Partial CBB Scores differed by 0.038 points (0.671 vs 0.709) across models.
- Critically, 100% confidence and 0.000 Error Recall persisted across ALL models tested, confirming the metacognitive blindspot is a cross-model structural phenomenon, not a single-model artifact.

Cross-Model Finding: The overconfidence and error forecasting failure are not unique to one model. They appear consistently across model capability levels, suggesting a systemic training-level issue rather than a model-specific quirk.

5.8.1. Full Leaderboard: 8 Models, 6 Vendors

CBB was subsequently tested on the Kaggle Benchmarks platform across 8 models from 6 vendors. Results confirmed clear discriminatory power:

Model	Vendor	Score	Status
Claude Sonnet 4.6	Anthropic	1.00	PASS 2/2
Gemini 2.5 Flash	Google	1.00	PASS 2/2
Gemini 2.0 Flash	Google	1.00	PASS 2/2
Gemma 3 27B	Google Open	1.00	PASS 2/2

Model	Vendor	Score	Status
GPT-5.4 mini	OpenAI	1.00	PASS 2/2
GLM-5	Z.ai	1.00	PASS 2/2
Qwen 3 Next 80B Instruct	QwenLM	1.00	PASS 2/2
DeepSeek-R1	DeepSeek	0.00	FAIL 1/2

DeepSeek-R1 was the only model to fail, suggesting its reasoning training creates a distinct metacognitive failure mode consistent with construction compulsion patterns reported in related work. 7 of 8 frontier models passed, confirming CBB correctly identifies well-calibrated models while discriminating against models with structural metacognitive deficits.

5.9 Failure Profile Taxonomy

Profile	Accurate?	Calibrated ?	Error Recall	Deployment Risk
Ideal	High	Yes	> 0	Minimal
Overconfident Correct (7 of 8 models)	High	No	0.000	Moderate - no self-flagging
Calibrated Incorrect	Low	Yes	> 0	Manageable - honest uncertainty
Overconfident Incorrect	Low	No	0.000	Critical - wrong and certain

All tested models fall into the Overconfident Correct profile. Standard benchmarks would rate these models highly based on accuracy alone. CBB reveals the consistent hidden risk profile across the capability spectrum.

6. Organizational Affiliations

None. Independent submission.

7. References & Citations

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8. Author & Project Links

Emmanuel Fle Chea, MPH - Clinical Data Scientist | Healthcare Data Scientist & Analyst | AI Systems Evaluation | Cognitive Science of Uncertainty | LLM Behavioral Analysis

- Kaggle Writeup: https://www.kaggle.com/competitions/kaggle-measuring-agi/writeups/calibration-blindspot-benchmark-cbb
- Kaggle Notebook: https://www.kaggle.com/code/efc04291993/cbb-kaggle-notebook
- GitHub: https://github.com/efchea1/calibration-blindspot-benchmark
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